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Systematic review of diet quality indices and their associations with health-related outcomes in children and adolescents

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diet index, diet quality, diet variety, nutrition assessment, pediatric, systematic review.

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Abstract

Background: Diet quality indices add an important dimension to dietary assessment. The aim of this systematic review was to: (i) identify and describe the attributes and applications of diet quality indices developed for use or used in paediatric populations; (ii) describe associations between these diet quality indices and health-related variables in paediatric populations; and (iii) identify factors that are associated with diet quality in paediatric populations worldwide.
Methods: Studies were identified by searching electronic databases for relevant papers from 1980 to October 2013 using keywords. Inclusion criteria were original studies that utilised a quantitative measure of diet quality in children and adolescents aged 0–18 years.
Results: One hundred and nineteen studies met the inclusion criteria, from which 80 different diet quality indices were identified. The majority of studies had >1000 participants and were of acceptable quality. Of the 56 studies that investigated health-related outcomes, weight status was the most researched. Europe produced the most number of diet quality indices (n = 27 indices). Of the 119 studies, seven intervention studies were identified. Paediatric diet quality indices were found to be associated with environmental, behavioural and maternal factors.
Conclusions: The use of diet quality indices in paediatric populations is a rapidly expanding area of research in diverse populations internationally. In economically disadvantaged countries, diet quality indices may be predictive of child growth. However, prospective cohort, intervention and validation studies are required to draw stronger conclusions concerning risk of future disease in paediatric populations in general.

Introduction

Assessment of diet quality is concerned with both the quality and variety of the holistic diet, rather than individual nutrients, and allows for the evaluation of how closely eating patterns align with dietary recommendations. In adults, diet quality has been used to quantify the risk of chronic conditions, including cardiovascular disease, some cancers, and total and disease specific mortality (Wirt & Collins, 2009; Kimokoti & Millen, 2011). Diet quality indices, or scores, are tools that provide an overall rating, on a numeric scale, of an individual's dietary intake in reference to nutrient and/or dietary recommendations (Wirt & Collins, 2009). The development and application of these tools is expanding rapidly, as is the exploration of relationships between indicators of health and disease (Kourlaba & Panagiotakos, 2009; Roman-Vinas *et al.*, 2009). To date, most adult indices have been developed for epidemiological purposes, although their clinical application has been encouraged (Wirt & Collins, 2009). Despite this, their utility in paediatric populations worldwide is relatively under studied.

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The assessment of paediatric diet quality is of high interest because food habits and behaviours that develop in childhood can track over time and predict adult diet-related disease (Patterson *et al.*, 2009; Craigie *et al.*, 2011). A number of paediatric diet quality indices have emerged, such as the North American Youth Healthy Eating Index (YHEI) (Feskanich *et al.*, 2004), the Revised Children's Diet Quality Index (RC-DQI) (Kranz *et al.*, 2008), the Australian Child and Adolescent Recommended Foods Score (ACARFS) (Marshall *et al.*, 2012) and the Greek Preschool Diet-Lifestyle Index (Manios *et al.*, 2010). However, an examination of all reported paediatric diet quality indices, including those in both developed and developing nations, their association with health-related outcomes, and a thorough exploration of factors associated with diet quality in children, remains incomplete. Two recent reviews have explored diet quality measures in paediatric populations from developed countries only. The review by Smithers *et al.* (2011) concluded that the use and application of diet quality indices, especially in regard to weight status in developed nations, had not been fully explored. A review by Lazarou & Newby (2011) also concluded that further research was required to gain a better understanding of paediatric diet quality and health-related outcomes. The present study expands on information presented in these reviews by incorporating populations from developing countries and additional studies published after 2010, as well as providing a more thorough summary of the diet quality indices' attributes and uses. This systematic review intends to be a complete summary and reference guide for all paediatric diet quality indices published up to October 2013. The specific aims of this review were to: (i) identify and describe the attributes and applications of diet quality indices developed for use or used in paediatric populations; (ii) describe associations between these diet quality indices and health-related variables in paediatric populations; and (iii) identify factors that are associated with diet quality in paediatric populations worldwide.

Materials and methods

A systematic literature review of current evidence was conducted in reference to the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) statement (Liberati *et al.*, 2009).

Search strategy

Published English-language studies were searched using the electronic databases PubMed and CINAHL (via Ebscohost) for publications from January 1980 to 31 October 2013. The search strategy was conducted using the keywords: (*pediatric* OR *child** OR *paed**) AND (*diet quality* OR *diet variety* OR *food variety* OR *diet** *diversity*).

The search strategy was complemented by a 'snowball' search, where retrieved papers had their reference lists hand searched to ensure that all relevant studies were identified. Variables and factors of interest included: (i) type of index (food or nutrient based); (ii) diet quality index country of origin; (iii) dietary method used by diet quality index; (iv) purpose of diet quality index; (v) method of scoring; (vi) age group appropriate for diet quality index; (vii) health-related outcomes associated with a diet quality index; and (viii) nonhealth-related factors found to impact upon diet quality.

Inclusion criteria for types of participants were children and/or adolescents aged 0–18 years in any setting. If adults were part of the sample population, the studies were included but only results for those aged ≤18 years were investigated. If the age range was not reported, samples were included if the mean age was ≤18.0 years. Inclusion criteria for types of studies were methodological studies, cohort studies, cross-sectional studies, case–control studies or experimental studies (e.g. randomised controlled trials, pre-post trials). Abstracts and full texts were considered and studies were included only if they described the development, validation, application or evaluation of an assessment tool to quantitatively measure diet quality and/or variety and/or diversity. Exclusion criteria for participants were adult populations (>18 years) and animal studies. Exclusion criteria for types of studies included review studies, non-English language publications, and studies that did not apply or describe a dietary scoring system. No exclusions were made based on variables of interest, for example, otherwise eligible papers that did not report association with a health-related variable were included, so that a diet quality index may still be identified and described.

Selection of studies and data synthesis

A two-step screening process was employed. In step 1, one author scanned the titles and abstracts of studies identified by the search for their eligibility and duplicates were removed. In step 2, full-text articles were screened by one author for eligibility and were confirmed by a second author if eligibility was unclear. Data were extracted from the published papers into standardised tables by two authors. The study design and population were reported in tables only for studies reporting the results for health-related variables. Health-related variables and non-health-related factors that may impact upon diet quality were considered significant at $P < 0.05$ and no

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exclusions were made for the type of statistical approach. All variables of interest were reported in tables, except for nonhealth-related factors, which were described in the text.

Review of study quality

Study methodological quality was assessed using the standardised critical appraisal instrument from the Joanna Briggs Institute (Schultz & Florence, 2007). Nine quality questions were classified as 'yes', 'no' or 'unclear' in relation to adequacy of the study sample description, use of objective outcome measures, adequacy of follow-up, appropriate statistical analysis and consideration of confounding factors. Quality data were extracted into standardised tables by one investigator and checked for completeness and accuracy by a second.

Results

Three hundred and twenty-four full-text studies were identified and retrieved for evaluation against the inclusion criteria, 119 of which were included (Fig. 1). Major reasons for exclusion were no use of a diet quality index (n = 131) and a non-paediatric population (n = 54). Of the 119 included studies, 80 different diet quality indices were identified that had been designed for use, or had been used, in paediatric populations. The sample size across studies ranged from nine to 165 111 and the majority had >1000 participants. Food frequency questionnaires and 24-h recalls were the most common dietary method; however, some indices required multiple dietary methods to be used to derive a score (e.g. 24-h recall and diet history). The age group appropriate for the indices was not always specified (Table 1). Studies were more likely to be published after 2005 (n = 89 studies), with 44 published from January 2010 and none before 1995. Fifty-six of the 119 included studies investigated associations between a diet quality index and health-related variables, using 40 of the 80 paediatric diet quality indices (see supporting information, Table S1).

The majority of included studies were rated as acceptable study quality and utilised appropriate statistical approaches (data not shown) (Schultz & Florence, 2007). Many of the studies utilised the same dataset; for example, data from the US Department of Agriculture Continuing Survey of Food Intakes by Individuals was used in seven studies (Kennedy *et al.*, 1995; Griel *et al.*, 2004; Kranz *et al.*, 2004, 2006; Knol *et al.*, 2005; Goodwin *et al.*, 2006; Beydoun & Wang, 2009). There were only seven intervention studies, one of which was a pilot study (Alexy *et al.*, 1999; Dixon *et al.*, 2000; Gittelsohn *et al.*, 2010; Vitolo *et al.*, 2010; Laster *et al.*, 2013; Rauber *et al.*, 2013; Spence *et al.*, 2013). The majority of the observational studies reported results cross-sectionally (see Supporting information, Table S1).

Table 1 describes the diet quality indices (n = 80 total) and includes tools developed in North America (n = 15), Europe (n = 27), Asia-Pacific (n = 17), Latin America (n = 8), Africa (n = 12) and one of unknown origin. Although an index was developed for a specific population, many have been subsequently used in diverse populations, with or without modifications. For example, several unevaluated modifications to the Health Eating Index (HEI) subscales were made for culturally diverse populations and specific geographical locations, including Brazil (de Andrade *et al.*, 2009; Vitolo *et al.*, 2010), Spain (Royo-Bordonada *et al.*, 2003a) and for Native Hawaiians in the USA (Gittelsohn *et al.*, 2010). In addition, many of the indices are updated or reappropriated versions of an

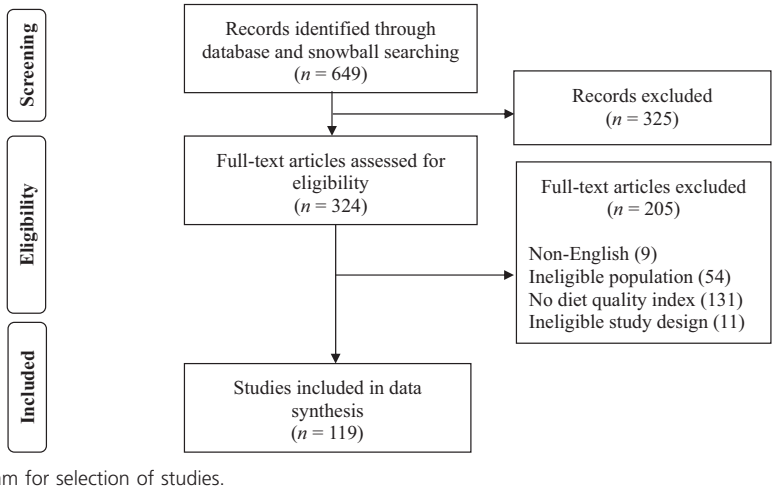


Figure 1 Study flow diagram for selection of studies.

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Table 1 Description and purpose of diet quality indices which have been designed for use or used in paediatric populations (n = 80 indices)

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
1	Healthy Eating Index (HEI); Kennedy *et al.* (1995)	Food and nutrient	USA	(i) 24-h recall and food diary/record; (ii) FFQ; (iii) 24-h recall; (iv) Food diary/ record	Reflects adherence to the Dietary Guidelines for Americans and the USDA Food Guide Pyramid (1992)	10 components, scored 0–100, calculated by nutrient analysis	≥2 years
2	Healthy Eating Index-2005 (HEI-2005); Britten *et al.* (2006); Guenther *et al.* (2008)	Food and nutrient	USA	(i) 24-h recall; (ii) food diary/record	Reflects adherence to the 2005 My Pyramid Food Guidance System	12 components, scored 0–100, calculated by nutrient analysis	2–18 years
3	Youth Healthy Eating Index (YHEI); Feskanich *et al.* (2004)	Food and behaviour	USA	(i) FFQ	Reflects adherence to the Dietary Guidelines for Americans	13 components, scored 0–100, points awarded and summed	Not specified
4	Canadian Healthy Eating Index (HEI-C); Glanville & McIntyre (2006)	Food and nutrient	Canada	(i) 24-h recall; (ii) 24-h recall and FFQ	Reflects adherence to the 1993 Canada's Good Guide to Healthy Eating and 1990 Canadian Nutrient Recommendations	Nine components, scored 0–100, calculated by nutrient analysis and servings	≥3 years
5	Canadian Health Eating Index-2009 (HEIC-2009); Woodruff & Hanning (2010)	Food and nutrient	Canada	(i) 24-h recall and FFQ	Reflects adherence to the 2007 Eating Well with Canada's Food Guide	Nine components, scored 0–100, calculated by nutrient analysis and servings	≥3 years
6	Alternative Health Eating Index (AHEI); McCullough *et al.* (2002)	Food and nutrient	USA	(i) FFQ	Reflects dietary patterns to lower risk of chronic disease	Nine components, scored 2.5–87.5, calculated by nutrient analysis and servings	Designed for adults, used in children
7	Diet Quality Index (DQI); Patterson *et al.* (1994)	Food and nutrient	USA	(i) 24-h recall; (ii) 24-h recall and food diary/record	Reflects adherence to the 1989 Recommended Dietary Allowances	Eight components, scored 0–16, points awarded and summed	Designed for adults, used in children
8	Diet Quality Index-International (DQI-I); Kim *et al.* (2003)	Food and nutrient	Designed in USA to be of international use	(i) FFQ; (ii) 24-h recall and FFQ	Reflects worldwide (WHO, USA and China) adherence to dietary food and nutrient recommendations	Four major components with sub-components, scored 0–100, calculated by nutrient analysis and servings	Not specified
9	Children's Diet Quality Index (C-DQI); Kranz *et al.* (2004)	Food and nutrient	USA	(i) 24-h recall	Reflects adherence to the 1998 Food GuidePyramid for 2–6 years old for components relevant to public health	Eight components, scored 0–70, calculated by nutrient analysis and servings	2–5 years

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Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
10	Revised Children's Diet Quality Index (RC-DQI); (Kranz *et al.* (2006)	Food, nutrient and behaviour	USA	(i) 24-h recall; (ii) Food diary/record	Reflects adequacy of nutrients and food group intakes which are of a public health concern	13 components, scored 0–95, calculated by nutrient analysis and servings	2–18 years
11	Food Variety Index for Toddlers (VIT); Cox *et al.* (1997)	Food	USA	(i) 24-h recall and food diary/record; (ii) Food diary/record	Reflects meeting the minimum requirement of the 1992 Food Guide Pyramid food groups.	Five components considered, scored 0.00–1.00, food items summed and truncated at 33%	Toddlers
12	Grain, Fruit, Vegetables, Dairy and Mild (GFVDM) Variety Score; Falciglia *et al.* (2004)	Food	USA	(i) 24-h recall	Reflects adherence to the 1992 Food Guide Pyramid food groups	Five food categories, servings summed	Not specified
13	Grain, Fruit and Vegetable (GFV) Variety Score; Falciglia *et al.* (2004)	Food	USA	(i) 24-h recall	Reflects the 2000 Dietary Guidelines for variety	Three food categories, servings summed	Not specified
14	Nutrient Rich Foods (NRF) 9.3; Fulgoni *et al.* (2009)	Nutrient	USA	(i) 24-h recall	Reflects nutritional adequacy	12 nutrients, scored <0–100, calculated via nutrient analysis	4 years – adult
15	Food Variety (FV) Score; Zimmer *et al.* (2012)	Food	USA	(i) FFQ	Reflects food variety	174 food items, scored 0–175, serves summed	Children
16	Diet Quality Index for Preschool Children (DQI-CH); (Huybrechts *et al.* (2010)	Food and behaviour	Belgium	(i) Food diary/record or FFQ	Reflects compliance with Flemish Food-Based Dietary Guidelines	Four components, scored −25 to 100%, calculated as percentage of food group intakes	2–6 years
17	Excess Index; Sabbe *et al.* (2008)	Food	Belgium	(i) FFQ; (ii) FFQ and food diary/record	Reflects weekly consumption of sugar- &/or fat-containing products	Number of items not specified, scored 7–35, servings summed	Not specified
18	Daily Diversity Index (DDI); Sabbe *et al.* (2008)	Food	Belgium	(i) FFQ	Reflects adherence to the 2000 American Food Guide Pyramid	Five components, scored 0–5, points awarded and summed	Not specified
19	Variety Index; Vereecken *et al.* (2008)	Food	Belgium and Italy	(i) FFQ and food diary/record	Reflects consumption of various food items	Seven components, scoring not specified	Not specified
20	Fruit and Vegetable Index (FV Index); Vereecken *et al.* (2008)	Food	Belgium and Italy	(i) FFQ and food diary/record	Reflects consumption of fruit and vegetable intake	No components specified, scoring not specified	Not specified
21	Fibre Index; Vereecken *et al.* (2008)	Food	Belgium and Italy	(i) FFQ and food diary/record	Reflects consumption of the fibre rich food groups fruit, vegetables and brown bread	No components specified, scoring not specified	Not specified

Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
22	Calcium Index (Ca-Index); Vereecken *et al.* (2008)	Food	Belgium and Italy	(i) FFQ and food diary/record	Reflects consumption of dairy items	Four components, scoring not specified	Not specified
23	Healthy Diet Indicator (HDI); Huijbregts *et al.* (1997)	Food and nutrient	Finland, Italy and Netherlands	(i) Diet history; (ii) food diary/record	Reflects WHO guidelines to prevent chronic disease	Nine components, scored 0–9, points awarded and summed	Designed for adults, used in children
24	Food-Variety Score (FVS); Emond *et al.* (2010)	Food	England	(i) FFQ	Reflects variety of foods consumed in a given FFQ	56 food items, scored 0–56, points awarded and summed	Not specified
25	Complementary Feeding Utility Index (CFUI); Golley *et al.* (2012)	Food and behaviour	England	(i) FFQ and independent questionnaire	Reflects adherence to commentary feeding guidelines in Australia, New Zealand, USA and UK	14 components, scored 0–1, calculated by summing of probability functions	Infants and toddlers
26	Healthy Food Diversity (HFD) – Index; Drescher *et al.* (2007)	Food	Germany	(i) FFQ; (ii) diet history	Reflects diversity and adherence to the Dietary Guidelines for an OMD for Adults	18 categories, scored 0–1/ 1- *n*, calculated by summing health factors	Designed for adults, used in adolesceents
27	Healthy Nutrition Score for Kids and Youth (HuSKY); Kleiser *et al.* (2009)	Food	Germany	(i) FFQ	Reflects adherence to the Dietary Guidelines for an OMD for Children and Adolescents	11 components, scored 0–100, calculated as ratio of food group intakes	3–17 years
28	Indicator Food Index (IFI); Kleiser *et al.* (2007)*	Food	Germany	(i) FFQ	Reflects adherence to the Dietary Guidelines for an OMD for Children and Adolescents	Seven food groups, scored 0 –14, points awarded and summed	Children and Adolescents
29	Recommended Food Group Change Score (RFS); Alexy *et al.* (1999)	Food	Germany	(i) Food diary/record	Reflects average change in amounts of deviation from the OMD food groups based on individualised recommendations	Number of components change depending on individualised recommendations, scored as a negative or positive percentage of change, servings summed	Not specified
30	Total Food Group Change Score (TFS); Alexy *et al.* (1999)	Food	Germany	(i) Food diary/record	Reflects average change in amounts of deviation from all OMD food groups	11 components, scored as a negative or positive percentage of change, servings summed	Not specified
31	Nutrient Improvement Score (NIS); Alexy *et al.* (1999)	Nutrient	Germany	(i) Food diary/record	Reflects average change in dietary intake of German reference values for nutrient intakes	16 components, scored as a '–' or '+' percentage of change, scored by nutrient analysis	Not specified

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Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
32	Nutrition Quality Index (NQI) Gedrich & Karg (2001)†	Nutrient	Germany	(i) Food diary/record	Reflects adequacy as compared the 2002 German, Austrian and Swiss Dietary Reference Values	13–17 components, scored 0–100, calculated by nutrient analysis	Not specified
33	Diet Quality Score; Kohlboeck *et al.* (2012)	Food	Germany	(i) FFQ	Reflects adherence to the OMD food groups based	11 components, scored 0–11, points awarded and summed	Children
34	Mediterranean Diet Score (MDS); Trichopoulou *et al.* (1995)	Food and nutrient	Greece	(i) FFQ; (i) food diary/ record	Reflects adherence to the Mediterranean dietary pattern	Eight components, scored 0–8, points awarded and summed	Designed for adults, used in children
35	Preschool Diet-Lifestyle Index (PDL-Index); Manios *et al.* (2010)	Food and behaviour	Greece	(i) 24-h recall and food diary/record	Reflects adherence to American and Canadian diet and physical activity guidelines for preschoolers	11 components, scored 0–44, points awarded and summed	2–5 years
36	Unhealthy Food Choices Score (UFCS); Yannakoulia *et al.* (2004)	Food	Greece	(i) FFQ	Reflects adherence to a number of Greek and international recommendations	Nine categories, scored 9–45, negative and positive points awarded and summed	11–15 years
37	E-KINDEX; Lazarou *et al.* (2008)	Food and behaviour	Greece	Unclear	Reflects risk of being overweight or obese.	Composed of three indices with a total of 30 components, scored 1–87	Not specified
38	Foods E-KINDEX; Lazarou *et al.* (2009a)	Food and behaviour	Cyprus	(i) FFQ	Score is one index used in the calculation of the E-KINDEX	13 components, scored 0–37, unspecified scoring method	Not specified
39	Diet Quality Score; Crombie *et al.*(2009)	Food	Scotland	Unclear	Reflects adherence to the Caroline Walker Trust recommendations for under 5s	Five components, dichotomous scoring for each component summed	2–5 years
40	Diet Variety Index (DVI); Royo-Bordonada *et al.* (2003b)	Food	Spain	(i) FFQ	Indicates diet variety	75 food items, scored 0–75, servings summed	Not specified
41	Mediterranean Diet Quality Index International (Med DQI-I); Mariscal-Arcas *et al.* (2007)	Food and nutrient	Spain	(i) 24-h recall and FFQ	Reflects worldwide (WHO, USA and China) adherence to dietary food and nutrient recommendations with specific Mediterranean adaptations	Four major components each with sub-components, scored 0–100, calculated by nutrient analysis	Not specified

Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
42	Mediterranean Diet Quality Index for children and adolescents (KIDMED); Serra-Majem *et al.* (2004)	Food and behaviour	Spain	(i) 24-h recall and FFQ; (ii) independent questionnaire; (iii) FFQ and independent questionnaire; (iv) 24-h recall	Reflects adherence to the Mediterranean Diet Model	16 components, scored 0–12, points awarded and summed	Not specified
43	Dietary Guideline Index for Children and Adolescents (DGI-CA); Golley *et al.* (2011)	Food	Australia	(i) 24-h recall	Reflects adherence to the 2003 Australian Dietary Guidelines for Children and Adolescents and 1998 Australian Guide to Healthy Eating	11 components, scored 0–100, calculation by nutrient analysis	Not specified
44	Australian Child and Adolescent Recommended Food Score (ACARFS); Marshall *et al.* (2012)	Food	Australia	(i) FFQ	Reflects adherence to the 2003 Australian Dietary Guidelines for Children and Adolescents	Eight components, scored 0–73, points awarded and summed	6–14 years
45	Diet Quality Index; Li *et al.* (2012)	Food and nutrient	Australia	(i) FFQ	Reflects adherence to the Australian Guide to Healthy Eating and Nutrient Reference Values for Australia and New Zealand	12 components, scored 20–150,	Adolescents
46	Core Food Variety Score (CFVS); Scott *et al.* (2012)	Food	Australia	(i) 24-h recall	Reflects adherence to the Australian Guide to Healthy Eating	Six food groups, scored 0–34, points awarded and summed	Toddlers
47	Fruit and Vegetable Variety Score (FVVS); Scott *et al.* (2012)	Food	Australia	(i) 24-h recall	Reflects adherence to the Australian Guide to Healthy Eating	Six food groups, scored 0–16, points awarded and summed	Toddlers
48	Obesity Protective Dietary Index (OPDI); Spence *et al.* (2013)	Food	Australia	(i) 24-h recall	Reflects adherence to a non-obesogenic diet	Three food groups, scored 0–30, unspecified scoring method	Infants
49	Adolescent Micronutrient Quality Index (AMQI); Chiplonkar & Tupe (2010)	Food	India	(i) 24-h recall	Reflects adherence to the 2005 Dietary Guidelines for Indians and the 2005 Dietary Guidelines for Americans	13 components, scored 0–100, points awarded and summed, unspecified scoring method	Adolescents

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Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
50	Diet Variety Score (DV Score) Mendez (2000)‡	Food	Philippines	(i) Independent questionnaire	Indicates diet variety	Eight components, scored 0–8, servings summed	Not specified
51	Dietary Diversity Score (DDS); Kennedy *et al.* (2007)	Food	Philippians	(i) 24-h recall; (ii) FFQ	Indicates diet diversity in adherence with development and analysis guidelines for developing countries Arimond *et al.* (2005), Kennedy & Nantel (2006)	10 components, scored 1–9, servings summed	Non-breastfed infants
52	Dietary Diversity Score 10 g (DDS 10 g); Kennedy *et al.* (2007)	Food	Philippians	(i) 24-h recall; (ii) FFQ	Indicates diet diversity in adherence with development and analysis guidelines for developing countries Arimond *et al.* (2005), Kennedy & Nantel (2006)	10 components, scored 1–9, servings summed	Non-breastfed infants
53	Food Variety Score (FVS); Saibul *et al.* (2009)	Food	Malaysia	(i) 24-h recall	Indicates diet variety	69 food items, scored 0–69, servings summed	Not specified
54	Food Variety Score (FVS); Darapheak *et al.* (2013)	Food	Cambodia	(i) 24-h recall	Indicates diet diversity	Seven components, scored 0–9, servings summed	Infants and toddlers
55	Dietary Diversity Score (DDS); Rah *et al.* (2010)	Food	Bangladesh	(i) FFQ	Indicates diet diversity	Nine components, scored 0–63, servings summed	<5 years
56	Dietary Diversity Score (DDS); Rani *et al.* (2010)	Food	India	(i) 24-h recall	Indicates diet diversity	13 components, scored 0–13, servings summed	Designed for adults, used in children
57	Food Variety Score (FVS) (Rani *et al.*, 2010)	Food	India	(i) 24-h recall	Indicates diet diversity	13 components, scored 0–78, servings summed	Designed for adults, used in children
58	Dietary Diversity Score (DDS); Mirmiran *et al.* (2004)	Food	Iran	(i) 24 r recall	Indicates diet diversity	Five components with 27 sub-components, scored 0–10, points awarded and summed	≥10 years
59	Dietary Diversity Score (DDS); Shatenstein *et al.* (1996)	Food	Gaza Strip	(i) FFQ	Indicates diet diversity	Four components, scored 0–4, points awarded and summed	12–17 years

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Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
60	Infant and Child Feeding Index (ICFI); Ruel & Menon (2002)	Food and behaviour	Designed in the USA for use in Latin America	(i) 24-h recall and FFQ; (ii) 24-h recall; (iii) 7 days recall	Reflects adequacy of the 1998 WHO and the 1999 Academy of Educational Development complementary feeding recommendations	Five components, scored 0–12, points awarded and summed	6–36 months
61	Dietary Diversity Indicator; Arimond & Ruel (2004)	Food	Designed in the USA for use in Southeast Asia, Latin America and Africa	(i) 7 days recall; (ii) FFQ	Indicates diet diversity	Seven components, scored 0–7, points awarded and summed	Infants
62	Food Diversity Index (FDI); Benefice *et al.* (2006)	Food	Bolivia	(i) 24-h recall and FFQ	Indicates diet diversity	Four components, scored 0–19, servings summed	Paediatrics and adults
63	Traditional Food Diversity Score (TFDS); Roche *et al.* (2008)	Food	Peru	(i) 24-h recall	Indicates diet diversity	No components specified, no scoring limit, servings summed	Paediatrics and adults
64	1 and 3 day Food Variety Score (FVS); Enneman *et al.* (2009)	Food	Guatemala	(i) 24-h recall	Indicates diet diversity	No components specified, no scoring limit, servings summed	Infants
65	USAID Dietary Diversity Score (DDS); Enneman *et al.* (2009)	Food	Guatemala	(i) 24-h recall	Indicates diet diversity in adherence with USAID guidelines Dewey *et al.* (2006)	Eight components, scored 0–8, servings summed	Infants
66	Cooking Pot Dietary Diversity Score (DDS); Enneman *et al.* (2009)	Food	Guatemala	(i) 24-h recall	Indicates diet diversity in adherence with the Guatemalan dietary guide translated	Six components, scored 0–6, servings summed	Infants
67	INCAP Papers Dietary Diversity Score (DDS); Enneman *et al.* (2009)	Food	Guatemala	(i) 24-h recall	Indicates diet diversity in adherence with INCAP protocol Bermudez *et al.* (2008)	25 components, scored 0–25, servings summed	Infants
68	Dietary Diversity Score (DDS8); Moursi *et al.* (2008)	Food	Madagascar	(i) 24-h recall	Indicates diet diversity in adherence with development and analysis guidelines for developing countries Arimond *et al.* (2005)	Eight components, scored 0–8, points awarded and summed	Infants aged ≥6 months

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Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
69	Dietary Diversity Score (DDS8-R); Moursi *et al.* (2008)	Food	Madagascar	(i) 24-h recall	Indicates diet diversity in adherence with development and analysis guidelines for developing countries Arimond *et al.* (2005)	Eight components, scored 0–8, points awarded and summed	Infants aged ≥6 months
70	Dietary Diversity Score (DDS7); Moursi *et al.* (2008)	Food	Madagascar	(i) 24-h recall	Indicates diet diversity in adherence with development and analysis guidelines for developing countries Arimond *et al.* (2005)	Seven components, scored 0–7, points awarded and summed	Infants aged ≥6 months
71	Dietary Diversity Score (DDS7-R); Moursi *et al.* (2008)	Food	Madagascar	(i) 24-h recall	Indicates diet diversity in adherence with development and analysis guidelines for developing countries Arimond *et al.* (2005)	Seven components, scored 0–7, points awarded and summed	Infants aged ≥6 months
72	Diet Diversity Score (DDS); Mpontshane *et al.* (2008)	Food	South Africa	(i) Diet history and food diary/record	Indicates diet diversity	Eight components, scored 0–100%, scored as a percentage of servings	Infants aged ≥6 months
73	Dietary Diversity Score (DDS); Hatloy *et al.* (1998)	Food	Designed in Norway to be of use in economically poor countries, such as Mali	(i) Food diary/record	Indicates diet diversity	Eight components, scored 0–8, points awarded and summed	13–58 months
74	Food Variety Score (FVS); Hatloy *et al.* (1998)	Food	Designed in Norway, used in economically poor countries	(i) Food diary/record; (ii) FFQ	Indicates diet diversity	No components specified, no scoring limit, servings summed	≥13 months to adults
75	Dietary Diversity Score (DDS); Torheim *et al.* (2004)	Food	Designed in Norway, used in economically poor countries	(i) FFQ	Indicates diet diversity	10 components, scored 0–10, points awarded and summed	≥15 years to adults
76	Dietary Diversity Score (DDS); Steyn *et al.* (2006)	Food	South Africa	(i) 24-h recall	Indicates diet diversity	Nine components, scored 0–9, points awarded and summed	1–8 years
77	Food Variety Score (FVS); Steyn *et al.* (2006)	Food	South Africa	(i) 24-h recall	Indicates diet diversity	No components specified, scored 0–45, servings summed	1–8 years

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Table 1 *(Continued)*

	Index and original author	Type of index	Country of origin	Dietary method	Purpose	Scoring	Age group
78	Dietary Adequacy Score (DAS); Guthrie & Scheer (1981)[§]	Food	Unknown	(i) 24-h recall; (ii) FFQ	Reflects adequacy of Recommended Dietary Allowances for an unknown population	Four components, scored 0–16, points awarded and summed	Not specified
79	Dietary Diversity Index (DDI); Bork *et al.* (2012)	Food	Senegal	(i) 24-h recall	Indicates diet diversity	Seven components, scored 0–7, points awarded and summed	Infants and toddlers
80	Food Variety Index (FVI); Bork *et al.* (2012)	Food	Senegal	(i) 24-h recall	Indicates food variety	20 components, scored 0–21, servings summed	Infants and toddlers

FFQ, food frequency questionnaire; INCAP, Institute of Nutrition of Central America and Panama; OMD, optimised mixed diet; USAID, United States Agency for International Development; USDA, United States Department of Agriculture; WHO, World Health Organisation.

*Kleiser *et al.* (2007) are cited as the original authors of the Indicator Food Index by Truthmann *et al.* (2012); however, the paper was not available in English.

[†]Gedrich & Karg (2001) are cited as the original authors of the Nutrition Quality Index by Libuda *et al.* (2009) & Cheng *et al.* (2010); however, the paper was not available for review.

[‡]Mendez (2000) is cited as the original author of the Diet Variety Score by Eckhardt *et al.* (2005); however, the thesis was not available for review.

[§]Guthrie & Scheer (1981) are cited as the original authors for the Dietary Adequacy Score by Shatenstein *et al.* (1996); however, the paper was not available for review.

original, for example the Diet Quality Index-International (DQI-I) (Kim *et al.*, 2003) and the Mediterranean DQI-I (Mariscal-Arcas *et al.*, 2007) were adapted from the original Diet Quality Index (Patterson *et al.*, 1994), whereas the RC-DQI (Kranz *et al.*, 2006) was updated from the Children's Diet Quality Index (Kranz *et al.*, 2004).

The majority of the paediatric diet quality indices were designed to assess the dietary intake of populations for epidemiological purposes. However, there were 11 indices that were described by their authors as being able to be applied to the assessment of an individual's dietary intake. These include the YHEI (Feskanich *et al.*, 2004), Healthy Nutrition Score for Kids & Youth (Kleiser *et al.*, 2009), Recommended Food Group Change Score (RFS), Total Food Group Change Score (TFS) and Nutrient Improvement Score (NIS) (Alexy *et al.*, 1999), the PDL- Index (Manios *et al.*, 2010), the E-KINDEX (Lazarou *et al.*, 2008), the Foods E-KINDEX (Lazarou *et al.*, 2009a), the Mediterranean Diet Quality Index for Children & Adolescents (KIDMED) (Serra-Majem *et al.*, 2004), the ACARFS (Marshall *et al.*, 2012) and the Adolescent Micronutrient Quality Index (AMQI) (Chiplonkar & Tupe, 2010).

Diet quality indices in developing nations

With the exception of the Adolescent Micronutrient Quality Index and the Infant and Child Feeding Index, paediatric diet quality in developing countries has been assessed using a range of dietary diversity scores or food variety scores (n = 31 scores). Because indices in developing nations have not been reviewed previously, a summary of their scoring and evaluation methodology is provided in Table 2. These tools were generally not named, nor developed to assess adherence to predetermined dietary recommendations, with the exception of the Cooking Pot Dietary Diversity Score (Enneman *et al.*, 2009). The scoring of these indices were based on the assumption that a diet with adequate variety of food groups or food items was more likely to be nutritionally adequate (Hatloy *et al.*, 1998). A number of tools (Shatenstein *et al.*, 1996; Mpontshane *et al.*, 2008; Enneman *et al.*, 2009; Emond *et al.*, 2010) were designed or implemented with no assessment made of validity (Table 2). Food categories reported in these studies were often arbitrary, reflecting local judgement related to regional foods of importance (Mpontshane *et al.*, 2008) and, in some studies, no explanation was given regarding food categorisation (Eckhardt *et al.*, 2005; Rah *et al.*, 2010). Several diversity or variety scores (Steyn *et al.*, 2006; Kennedy *et al.*, 2007; Daniels *et al.*, 2009; Cabalda *et al.*, 2011) were designed in reference to development guidelines (Arimond *et al.*, 2005; Dewey *et al.*, 2006; Kennedy & Nantel, 2008).

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Table 2 Scoring methodology and evaluation method of diet diversity scores and food variety scores that have been used in paediatric populations of developing countries (n = 31 indices)

Index and author	Scoring method	Evaluation
Diet Variety Score (DV score); Mendez (2000)	Scored 0–8. One point awarded for consuming one food from each of eight food categories: fish, animal sourced foods, staple cereals, other starches, vegetables, fruit, beans and nuts, dairy	Nutrient adequacy
Diet Diversity Score (DDS); Kennedy *et al.* (2007)*	Scored 0–10. One point awarded for consuming at least one food from each of 10 food categories: cereals and tubers; meat, poultry and fish; dairy; eggs; pulses and nuts; vitamin A rich fruit and vegetables; other fruit; other vegetables; oils and fats; and other	Nutrient adequacy
Diet Diversity Score 10 g (DDS 10 g); Kennedy *et al.* (2007)	Scored 1–10. One point awarded for consuming at least 10 g of food from each of 10 food categories: cereals and tubers; meat, poultry and fish; dairy; eggs; pulses and nuts; vitamin A richfruit and vegetables; other fruit; other vegetables; oils and fats (10 g minimum not applied); andother	Nutrient adequacy
Food Variety Score (FVS); Saibul *et al.* (2009)	Scored 0–69. One point awarded for consuming each of 69 food items	Energy adequacy
Food Variety Score (FVS); Emond *et al.* (2010)	Scored 0–56. One point awarded for never consuming each of 56 food items	Nil
Diet Diversity Score (DDS); Rah *et al.* (2010)	Scored 0–63. One point awarded for consuming, over a week, a food from nine food groups: rice, lentils, green leafy vegetables, yellow/orange fruit, eggs, fish, chicken, meat and milk	Growth
Dietary Diversity Score (DDS); Rani *et al.* (2010)	Scored 0–13. One point awarded for consuming a food from each 13 food categories: starchy staples, legumes and nuts, dairy, other fruit and vegetables, vitamin C rich fruit, vitamin C rich vegetables, vitamin A rich dark green leafy vegetables, vitamin A rich deep yellow, orange and red vegetables, organ meats, eggs, small fish eaten whole with bones, other flesh foods and miscellaneous animal protein	Nutrient adequacy
Food Variety Score (FVS); Rani *et al.* (2010)	Scored 0–78. One point awarded for consuming each food item from each 13 food categories: starchy staples, legumes and nuts, dairy, other fruit and vegetables, vitamin C rich fruit, vitamin C rich vegetables, vitamin A rich dark green leafy vegetables, vitamin A rich deep yellow, orange and red vegetables, organ meats, eggs, small fish eaten whole with bones, other flesh foods and miscellaneous animal protein	Nutrient adequacy
Dietary Diversity indicator; Arimond & Ruel (2004)	Scored 0–7. One point awarded for consuming an item ≥3 days in a week from seven foods categories: starchy staples, legumes, dairy, meat, poultry and fish or eggs, vitamin A rich fruit and vegetables, other fruit and vegetables, foods made with oil or butter	Height-for-age *Z*-score
Food Diversity Index (FDI); Benefice *et al.* (2006)	Scored 0–19. One to four points awarded for consuming an item from four food groups: fish, meat and milk; cereals; fruits; tubers and plantains	Height-for-age z-score
Traditional Food Diversity Score (TFDS); Roche *et al.* (2008)	Unlimited scoring. One point awarded for each unique traditional (to the Peruvian Amazon) food consumed	Nutrient adequacy
1 and 3 day Food Variety Score; Enneman *et al.* (2009)	Unlimited scoring. One point awarded for each item consumed in 1 and 3 days	Nil
USAID Dietary Diversity Score (DDS); Enneman *et al.* (2009)	Scored 0–8. One point awarded for consuming an item from each of eight food categories: grains, roots, and tubers; legumes and nuts; dairy; flesh foods; eggs; vitamin A rich fruit and vegetables; other fruit and vegetables; fats and oils	Nil
Cooking Pot Dietary Diversity Score (DDS); Enneman *et al.* (2009)	Scored 0–6. One point awarded for consuming an item from each of six food categories: grains (legumes), cereals and potatoes; herbs and vegetables; fruits; meat; milk and milk products; sugar and fats	Nil
INCAP Papers Dietary Diversity Score (DDS); Enneman *et al.* (2009)	Scored 0–25. One point awarded for consuming an item from each of 25 food categories: corn, corn tortilla and corn gruel; corn tamales; beans; rice; bread; breakfast cereals; other cereals; milk, dairy products and eggs; meat, beef and pork; game meat; chicken and turkey; fish and shellfish; green leaves; green and yellow vegetables; other vegetables; potatoes, root crops and plantain; fruit; fats and oils; sugar; beverages; snacks; desserts; soups; other mixed dishes with meat; and miscellaneous	Nil

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Table 2 *(Continued)*

Index and author	Scoring method	Evaluation
Dietary Diversity Score (DDS8); Moursi *et al.* (2008)	Scored 0–8. One point awarded for consuming ≥1 g from each of eight food categories: grains, roots and tubers; legumes and nuts; dairy products; flesh foods; eggs; vitamin A rich fruit and vegetables; other fruit and vegetables; fats and oils	Nutrient adequacy
Dietary Diversity Score (DDS8-R); Moursi *et al.* (2008)	Scored 0–8. One point awarded for consuming ≥10 g from each of seven food categories: grains, roots and tubers; legumes and nuts; dairy products; flesh foods; eggs; vitamin A rich fruit and vegetables; other fruit and vegetables; and consuming ≥1 g from fats and oils food category	Nutrient adequacy
Dietary Diversity Score (DDS7); Moursi *et al.* (2008)	Scored 0–7. One point awarded for consuming ≥1 g from each of seven food categories: grains, roots and tubers; legumes and nuts; dairy products; flesh foods; eggs; vitamin A rich fruit and vegetables; other fruit and vegetables	Nutrient adequacy
Dietary Diversity Score (DDS7-R); Moursi *et al.* (2008)	Scored 0–7. One point awarded for consuming ≥10 g from each of seven food categories: grains, roots and tubers; legumes and nuts; dairy products; flesh foods; eggs; vitamin A rich fruit and vegetables; other fruit and vegetables	Nutrient adequacy
Dietary Diversity Score (DDS); Mpontshane *et al.* (2008)	Scored 0–100. Scored as a percentage of eight food categories consumed each day in a week: breast milk; formula milk; dairy products; vegetable protein; non-milk animal protein; snacks and sweet beverages; fruit and vegetables; complex carbohydrates	Nil
Dietary Diversity Score (DDS); Hatloy *et al.* (1998)	Scored 0–8. One point awarded for consuming each of eight food groups over 3 days: staples; vegetables; milk; meat; fish; egg; fruits; green leaves	Nutrient adequacy
Food Variety Score (FVS); Hatloy *et al.* (1998)	Unlimited scoring. One point awarded for each item consumed in 1 and 3 days	Nutrient adequacy
Dietary Diversity Score (DDS); Steyn *et al.* (2006)	Scored 0–9. One point awarded for consuming one food from nine food groups over 24 h: cereals, roots and tubers; vitamin A rich fruits and vegetables; other fruits; other vegetables; legumes and nuts; meat, poultry and fish; fats and oils; dairy; eggs	Nutrient adequacy
Food Variety Score (FVS); Steyn *et al.* (2006)	Unlimited scoring. One point awarded for consuming each food item from recommended nine food groups over 24 h: cereals, roots and tubers; vitamin A rich fruits and vegetables; other fruits; other vegetables; legumes and nuts; meat, poultry and fish; fats and oils; dairy; eggs.	Nutrient adequacy
Dietary Diversity Score (DDS); Torheim *et al.* (2004)	Scored 0–10. One point awarded for consuming each of 11 food groups over 7 days: cereals; legumes; oil and sugar; fruit; vegetables; meat; milk; fish; eggs; green leaves; and other	Nutrient adequacy
Dietary Diversity Score (DDS); Mirmiran *et al.* (2004)	Scored 0–10. Two points awarded for consuming ≥0.5 serve from each of five food categories: grains; vegetables; fruit; meat; dairy.	Nutrient adequacy
Dietary Diversity Score (DDS); Shatenstein, *et al.* (1996)	Scored 0–4. One point awarded for consuming one serve from each of four food categories: grains; dairy; protein (excluding dairy); fruit and vegetables	Nil
Dietary Adequacy Score (DAS); Guthrie & Scheer (1981)	Scored 0–16. One point awarded for consuming a serve from two food categories: grains; fruit and vegetables; and two points awarded for consuming a serve from two food categories: protein (excluding dairy) and dairy; with each food category capped at four points	Nutrient adequacy
Dietary Diversity Index (DDI); Bork *et al.* (2012)	Scored 0–7. One point awarded for consuming a serve from seven food categories: animal milk products; animal-based foods; cereals and tubers; pulses and nuts; fruit and vegetables; vitamin A–rich foods; and food with fat added	Height-for-age z-score
Food Variety Index (FVI); Bork *et al.* (2012)	Scored 0–21. One point for consuming a serve from 20 food items: fresh milk; powered milk; sour milk; fresh fish; dried or smoked fish; eggs; meat; groundnuts; other legumes; vegetables/leaves; fruit; vitamin A-containing food; tubers/roots; millet gruel; milk-based millet gruel; millet couscous; millet porridge; rice; fat-containing foods; and bread/biscuits; together with a category 'other foods'	Height-for-age z-score
Food Variety Score (FVS); Darapheak *et al.* (2013)	Scored 0–9. One point awarded for consuming a serve from seven food categories, capped at nine serves: staple foods; animal foods; milk products; green leafy/orange colour vegetables; pulses; oils/fats; seeds	Growth

*Not original author. Original paper not available.

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Diet quality indices associated with health-related outcomes

Weight status was the most commonly measured health-related outcome ($n = 34$ studies) in relation to a diet quality score (see Supporting information, Table S1), followed by height/length ($n = 17$ studies). Serum biomarkers were examined in seven studies, pre-existing disease states in six, blood pressure in four and waist circumference in four. Diarrhoea and body composition were considered in three studies each, whereas infections were considered in two studies. Outcomes that were examined in only one study include dental caries, mental health, health-related quality of life, puberty and fever. No type of diet quality index or scoring approach was able to better predict or find a significant relationship with a health-related outcome. However, studies that utilised food-based scores were more likely to test for associations with biomarkers and height/length, and food- and nutrient-based scores were more likely to test for associations with measures of adiposity.

Associations with weight status in populations at risk of over-nutrition were examined in 26 studies, with only eight finding a significant relationship (Feskanich *et al.*, 2004; Mirmiran *et al.*, 2004; Kranz *et al.*, 2008; Lazarou *et al.*, 2008; Chiplonkar & Tupe, 2010; Kontogianni *et al.*, 2010; Manios *et al.*, 2010; Vitolo *et al.*, 2010; Golley *et al.*, 2011; Jennings *et al.*, 2011). Generally, this relationship was inconsistent with studies finding negative and positive associations (see Supporting information, Table S1).

Poor diet quality was associated with earlier onset of puberty by 0.4 years ($P = 0.02$), an established risk factor for hormone-related cancers, all-cause mortality and insulin resistance (Cheng *et al.*, 2010). Three studies found weak negative associations with diastolic blood pressure (Lazarou *et al.*, 2009a,b; Golley *et al.*, 2013); however, the study by Truthmann *et al.* (2012) found a weak positive association with diastolic blood pressure using the Indicator Food Index. Malnutrition (kwashiorkor/marasmus) was not associated with diet diversity (Sullivan *et al.*, 2006; Lin *et al.*, 2007). Cross-sectionally, the pre-existing disease state of human immunodeficiency virus infection had a negative association with diet quality, as did autism in two studies; however, no association with autism or developmental delay was found in a later study (Mpontshane *et al.*, 2008; Emond *et al.*, 2010; Zimmer *et al.*, 2012; Graf-Myles *et al.*, 2013). No association with poor mental health (internalising disorder) was found in adolescents (McMartin *et al.*, 2012).

In developing nations where a risk of under-nutrition rather than over-nutrition was the major concern, diet quality indices were found to be positively associated with measures of child growth in 14 studies (Ruel & Menon, 2002; Arimond & Ruel, 2004; Mirmiran *et al.*, 2004; Eckhardt *et al.*, 2005; Benefice *et al.*, 2006; Sawadogo *et al.*, 2006; Kennedy *et al.*, 2007; Moursi *et al.*, 2009; Saibul *et al.*, 2009; Zhang *et al.*, 2009; Chiplonkar & Tupe, 2010; Rah *et al.*, 2010; Bork *et al.*, 2012; Darapheak *et al.*, 2013), although some studies reported no association. Furthermore, if the child experienced diarrhoea at the time of assessment, they were more likely to report poor diet diversity (Zhang *et al.*, 2009; Rah *et al.*, 2010).

Factors associated with diet quality

Several demographic factors were found to be associated with diet quality, these included socioeconomic status (Serra-Majem *et al.*, 2004; Torheim *et al.*, 2004; Kranz *et al.*, 2008; de Andrade *et al.*, 2009; Kleiser *et al.*, 2010; Golley *et al.*, 2011), age (Torheim *et al.*, 2004; Kranz *et al.*, 2008; de Andrade *et al.*, 2009; Kleiser *et al.*, 2009, 2010; Golley *et al.*, 2011), sex (Kleiser *et al.*, 2009, 2010), ethnic or migration background (Kranz *et al.*, 2008; Kleiser *et al.*, 2009, 2010), geographical location (Serra-Majem *et al.*, 2004; Simen-Kapeu *et al.*, 2010; Darapheak *et al.*, 2013) and the degree of urbanisation (Serra-Majem *et al.*, 2004; Kleiser *et al.*, 2010).

Multiple behaviours were also found to impact upon diet quality in children and adolescents, including beverage patterns (LaRowe *et al.*, 2007), portion sizes (Colapinto *et al.*, 2007), dieting as a result of weight concerns, the frequency of family dinners (Veugelers *et al.*, 2005; Woodruff *et al.*, 2010), eating in front of the television, as well as television viewing in general (Veugelers *et al.*, 2005; Liang *et al.*, 2009; Simen-Kapeu *et al.*, 2010; Sisson *et al.*, 2012), and meal skipping (Veugelers *et al.*, 2005; Woodruff *et al.*, 2008, 2010; Woodruff & Hanning, 2009). Low diet quality was also found to be associated with an increased likelihood of behavioural and emotional symptoms in children (Kohlboeck *et al.*, 2012). Specific eating environments were found to impact adversely on diet quality, such as purchasing or consuming meals outside the home, rather than eating food prepared at home from grocery store ingredients (Veugelers *et al.*, 2005; Woodruff & Hanning, 2009). Florence *et al.* (2008) and Golley *et al.* (2013) identified that diet quality was a predictor of academic performance and intelligence; however, the association was weakened when considering maternal intelligence as a confounder.

Maternal education and employment and parental diet quality was positively associated with the child's diet quality, and providing maternal education in regard to their children's dietary intake from birth significantly improved the child's diet quality at age 3–4 years but not at 7–8 years (Beydoun & Wang, 2009; Manios *et al.*, 2009;

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Vitolo *et al.*, 2010; Darapheak *et al.*, 2013; Laster *et al.*, 2013; Rauber *et al.*, 2013). However, mothers were found to significantly overestimate child diet quality, and maternal healthy eating knowledge was not predictive of child diet quality but rather the mothers' ability to translate that knowledge into behaviours such restricting sweet food intake (odds ratio = 21.63, 95% CI = 2.70–173.30) or promoting fruit consumption (odds ratio = 52.94, 95% CI = 1.09–7.95) (Crombie *et al.*, 2009; Kourlaba *et al.*, 2009).

Discussion

This systematic review provides a comprehensive summary of the diet quality indices that have been developed internationally to assess the overall healthfulness of dietary intakes in children and adolescents. The use of diet quality indices in paediatric populations is increasing internationally in highly diverse populations.

Significant variation existed in how indices were calculated. Methods where a score could be directly calculated from the dietary assessment method, such as simple addition or awarding of points, have less participant and researcher burden. The low burden indices may be used within a clinical or fieldwork setting, where they are able provide immediate client feedback. Examples of these low burden scores include the KIDMED or ACARFS in developed nations, and all diet variety or diversity scores in developing nations. By contrast, indices that require intensive dietary assessment or additional analysis before a final score could be derived have a higher participant and researcher burden. In some cases, such as nutrient-derived scores, these analysis are complex, requiring multiple mathematical models, making these scores only applicable for group-level research; for example, the HEI and Nutrient Rich Foods 9.3 (Table 1). Interestingly, the wide variation observed in scoring methodology and type of diet quality index did not impact upon the ability of indices to predict health-related outcomes.

Continental trends were observed in the type of scoring method. Diet quality indices developed in Europe were predominantly food-based, and include a number of simpler diet variety or diet diversity scores. Alternatively, North American derived diet quality indices were largely food- and nutrient-based scores requiring nutrient analysis for calculation. The dietary methods and scoring for diet quality indices in developing nations were found to be more simplistic compared to developed nations overall. This may be because application in these countries needs to be less time and resource intensive as a result of the fieldwork setting, and applicable to populations where food is often shared communally (Hatloy *et al.*, 1998). Overall, diet variety and diversity scores appear to predict positive child growth, and have previously been reported as sensitive indicators of food insecurity in developing countries, with an independent relationship between dietary diversity and dietary quality being reported in some studies (Ruel, 2003; Arimond & Ruel, 2004).

Determining the association between diet quality indices and health-related outcomes in children and adolescents is an important method for evaluating the usefulness of a diet quality index for particular settings. The associations are principally cross-sectional in nature and therefore diet quality indices should not yet be used to determine risk of disease. The conflicting associations with weight status in developed nations may be the result of a poor study design; for example, only one study considered the confounding effect of physical activity (Lazarou *et al.*, 2010). Beyond body weight, the relationship between diet quality and health-related outcomes has not been examined extensively in the published literature, with studies often only evaluating a single health-related variable at one time-point. This may be a result of paediatric diet quality indices being a new area of research, where many sample populations have been derived from ongoing larger studies and therefore are not designed for the purpose of examining diet quality and health. Additionally, selecting appropriate health and disease outcomes is more challenging in paediatric populations compared to adults.

Limitations and recommendations for further research

The present review may be limited by publication bias and relevant studies may have been missed if they did not report the examination of diet quality, diet variety, diet diversity or food variety in the title, abstract or keywords. Only two electronic databases were used; however, the two selected are the most relevant to the nutrition sciences and were supported by a thorough snowball search of citations. The body of research identified in the present review is limited by its observational and cross-sectional nature. Although several studies have used diet quality indices as outcome measures in intervention studies, no intervention studies have examined how paediatric diet quality indices may be used to improve the health of the paediatric populations, such as using self-assessment and feedback.

The overall paediatric diet quality research is limited by very few indices being supported by validation studies or evaluation in regard to nutritional adequacy. Although the methodologies of many indices are based on sound principles, such as adherence to dietary guidelines, conclusions about their results should be interpreted with caution because scores may not correlate beneficially with nutrient intakes or health outcomes of interest. Indices that have been shown to be associated with nutrient

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intakes, nutrient biomarkers and with internal validity and/or reliability studies lend more confidence to their results. In addition, the diet quality scores identified that were not developed specifically for children, such as the DQI, should also be interpreted with caution. Future diet quality studies in children and adolescents should utilise tools developed specifically for paediatric populations with consideration given to the purpose of the index. Importantly, these tools should be supported by validation studies.

As a result of the limitations in the existing body of knowledge, key areas for future research have been identified. Cohort studies that evaluate the relationship between paediatric diet quality and risk factors for adult chronic conditions, such as blood pressure and plasma lipids, are needed. Clarity is required for the association between diet quality indices and weight status in developed nations, where studies should employ several measures of anthropometry and use multivariate statistical models with all confounding variables considered. Examination between parent and child diet quality would complement research into chronic disease risk and provide evidence for more relevant recommendations and policies. Intervention studies in the clinical setting will extend paediatric diet quality index research beyond its established role in epidemiology.

Conclusions

The use of diet quality indices in paediatric populations is a rapidly expanding area of research in diverse populations internationally. Diet quality indices in paediatric populations have an established role in observational research and epidemiology and an emerging role in clinical application and intervention studies. Food behaviours and environmental and maternal influences impact upon a child or adolescents' diet quality and therefore future disease risk. In economically disadvantaged countries, diet quality indices may be predictive of child growth. However, prospective cohort, intervention and validation studies are required to draw stronger conclusions concerning the risk of future disease in paediatric populations in general.

Conflict of interests, source of funding and authorship

All authors declare that they have no conflicts of interest.
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SM carried out the literature search and was primarily responsible for data extraction and analysis, interpretation of data, as well as drafting and revision of the manuscript. TB reviewed the study quality and strength and contributed to data extraction, as well as drafting and revision of the manuscript. CC provided guidance and helped with revision of the manuscript. All authors critically reviewed the manuscript and approved the final version submitted for publication.

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Supporting information

Additional Supporting Information may be found in the online version of this article:
Table S1. Characteristics and results of studies examining associations between diet quality indices and health-related outcomes in paediatric populations (n = 56 of 119 studies).

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